Robust Operation

ECE 2564, Embedded Systems

Fall 2025

Overview

- Detecting brownout conditions
- Using watchdog timers

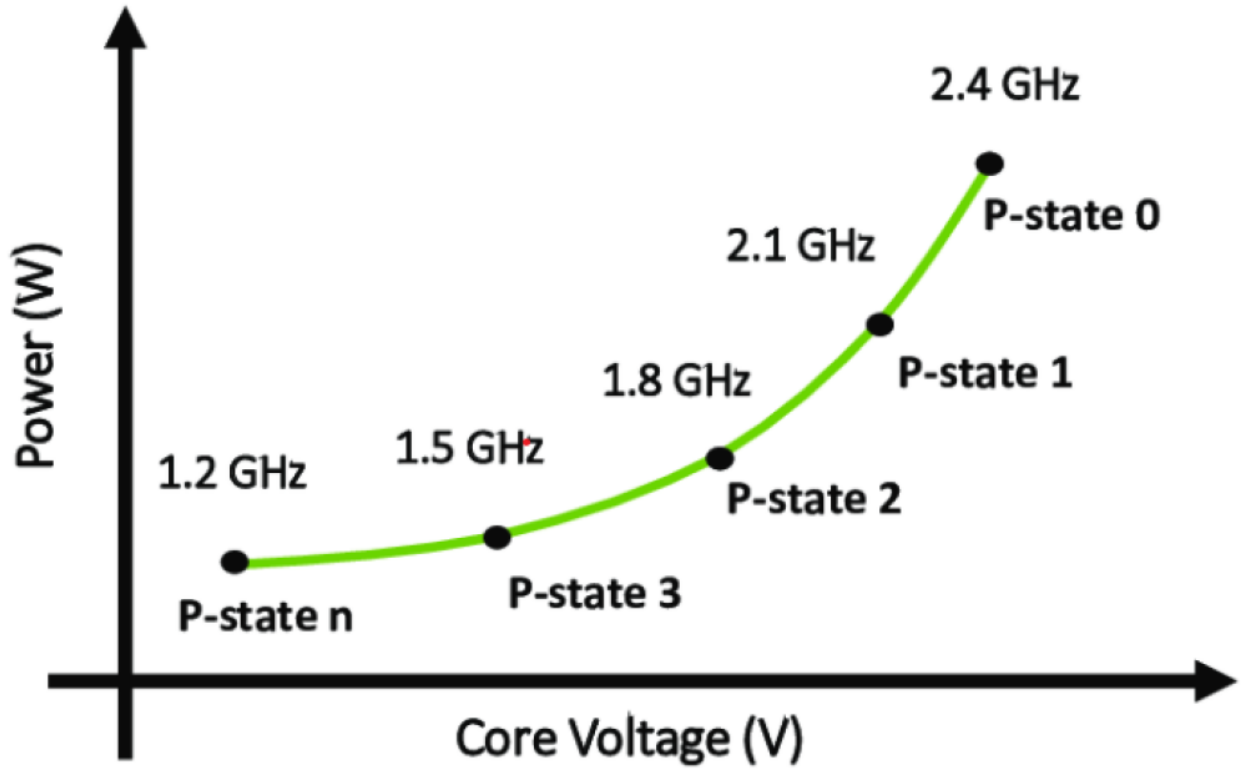
Import the repository **https://github.com/ECE2564-VT/basic_example_watchdogTimer.git**

Motivation for Robust Operation

- *Robustness* is the ability of a computer system to cope with erroneous input, and to cope with unexpected errors during execution
  - In some cases the best way to cope with bad situations is to reset the device and then "bring it back to life"
  - We'll consider some common mechanisms for doing this
- Terminology
  - Hard reset: Equivalent to cycling the power
  - Soft reset: Allow code to continue execution
- Brownout reset
  - A reset, possibly a soft reset, that is performed in response to low voltage from the power supply
- Watchdog timer

Voltage-Frequency Scaling

- What is the typical operating voltage of a circuit?
- It depends on the circuit and the speed you wish for it to run



Brownout Conditions

What happens when the supply voltage drops below the typical operating voltage?

Voltage drop	Example voltage fraction	Frequency	Power consumption	Operation
Negligible	0·95	3MHz	typical	normal
small	0·8	2MHz	lower	Power-saving mode
medium	0·5	1·8MHz	Even lower	Not good!! Mistakes, security attack
large	0·2	--	zero	Not working at all

MSP432 Brownout Control

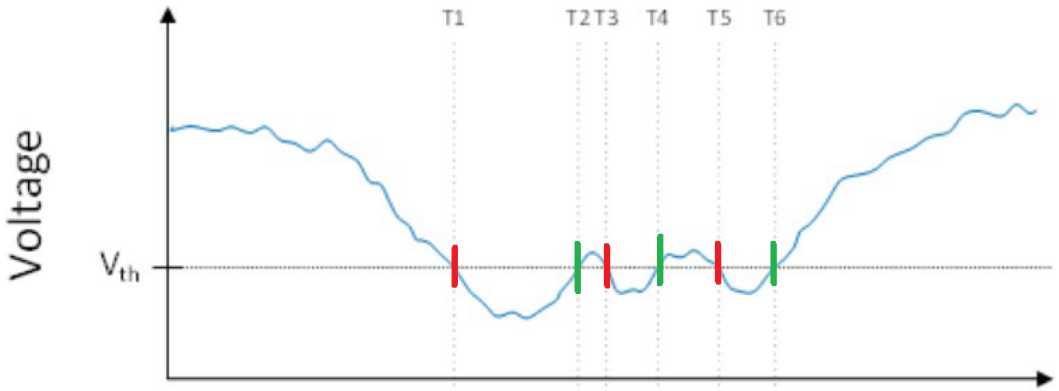
6.8.2 *Power Supply System (PSS)*

The PSS controls all the power supply related functionality of the device. It consists of the following components.

6.8.2.1 VCCDET

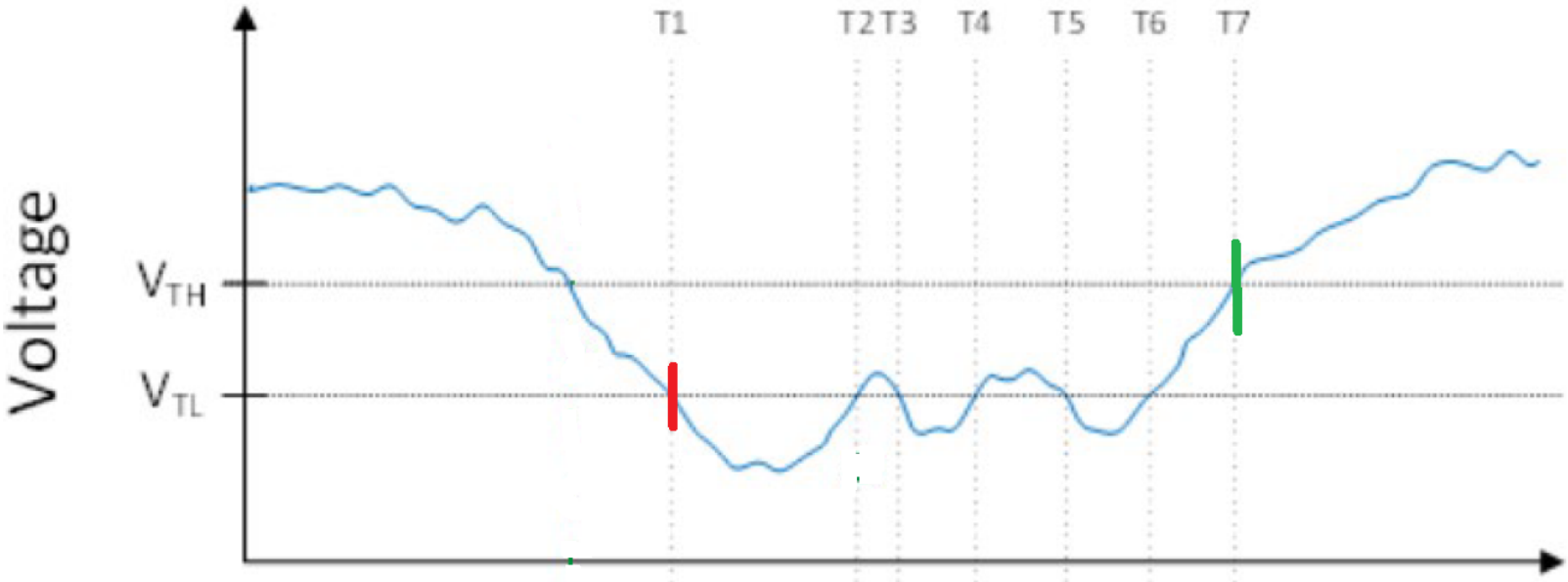
The VCCDET monitors the input voltage applied at the DVCC and AVCC pins of the device. When the V_{CC} is found to be below the operating range of the VCCDET trip points, it generates a brownout condition, thereby initiating a device reset (POR class reset).

- Suppose a circuit resets the MSP432 using a single threshold
- Between T1 and T2, T3 and T4, and T5 and T6, the CPU will turn off
- The brownout reset will trigger at T1, T3, and T5



Dual Threshold Brownout Control

- Consider the use of a *dual* threshold
  - The reset control now has hysteresis, similar to the functionality of the Schmitt-trigger device
  - The CPU shuts down when voltage drops below V_{TL} and will not turn on until the voltage rises above V_{TH}
  - The CPU shuts down at T1 and turns back on at T7



Brownout Control

- Most microprocessors and microcontrollers are designed with the ability to sense brownout conditions and perform a reset in response
- The reset is considered better than allowing the system to continue operating, when we expect unpredictable results

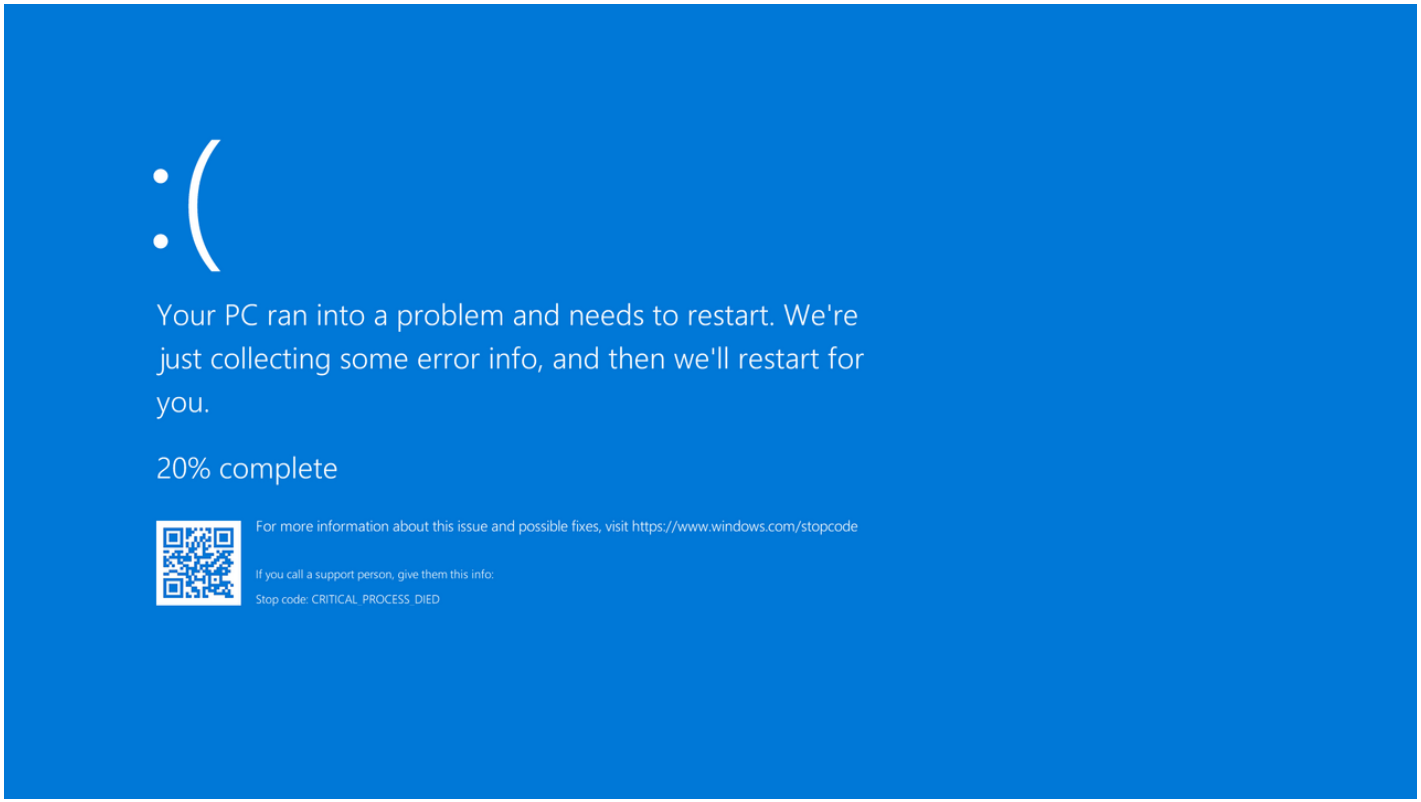
Watchdog Timer

- We've seen that MSP432 power management responds to brownout condition which are detected with circuitry.

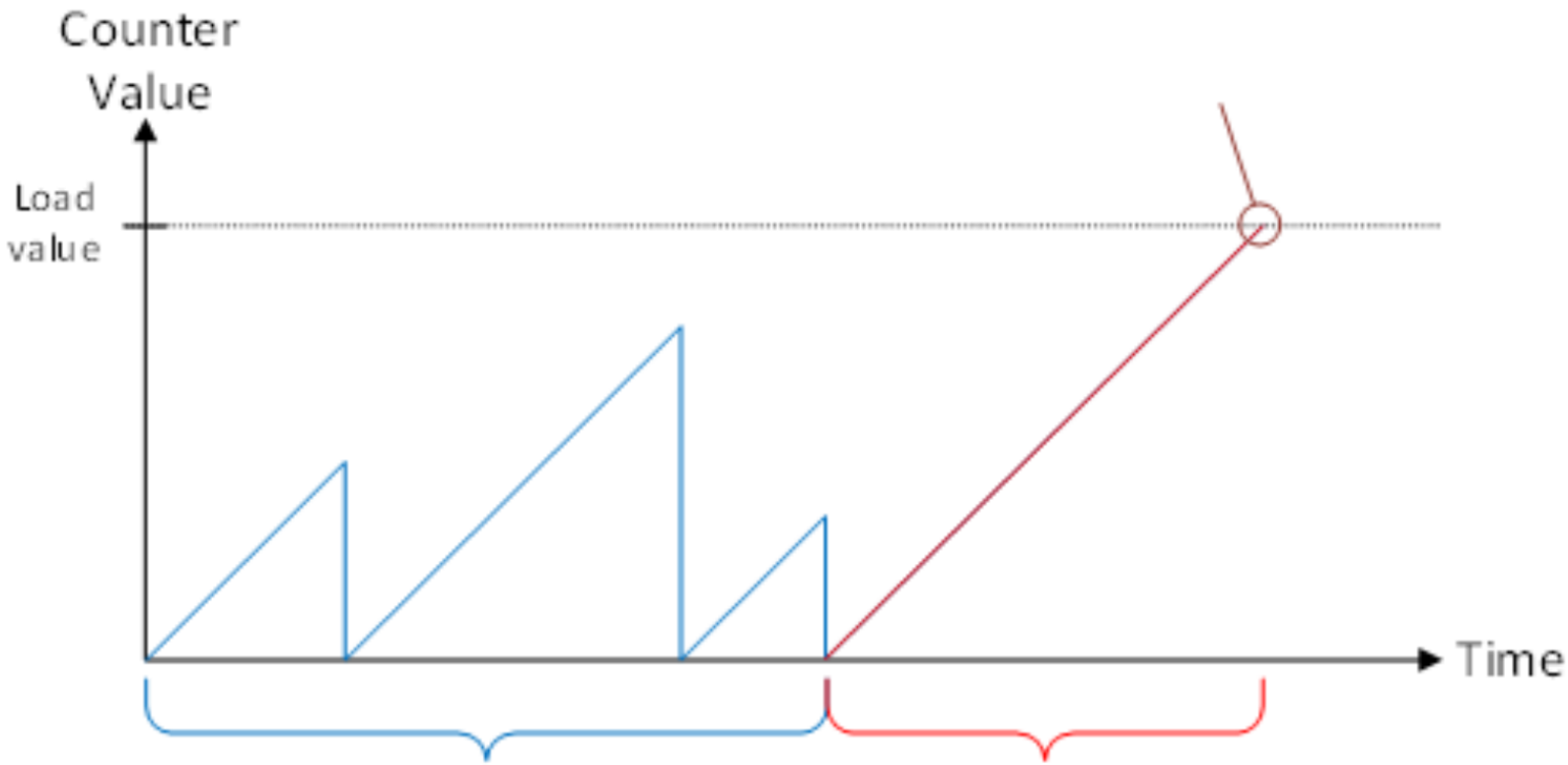
- Suppose the processor begins operating erratically for some other unknow reason
- We can use a *watchdog timer* to remedy this
- Key ideas:
  - The watchdog timer runs automatically and will reset the system if the timer ever expires
  - The developer writes software that frequently resets the watchdog timer

Watchdog Timer in Action

- Who has seen the Windows OS dreaded blue screen of death?
- It was triggered by the system's watchdog timer!



The Watchdog (Count-up) Timer



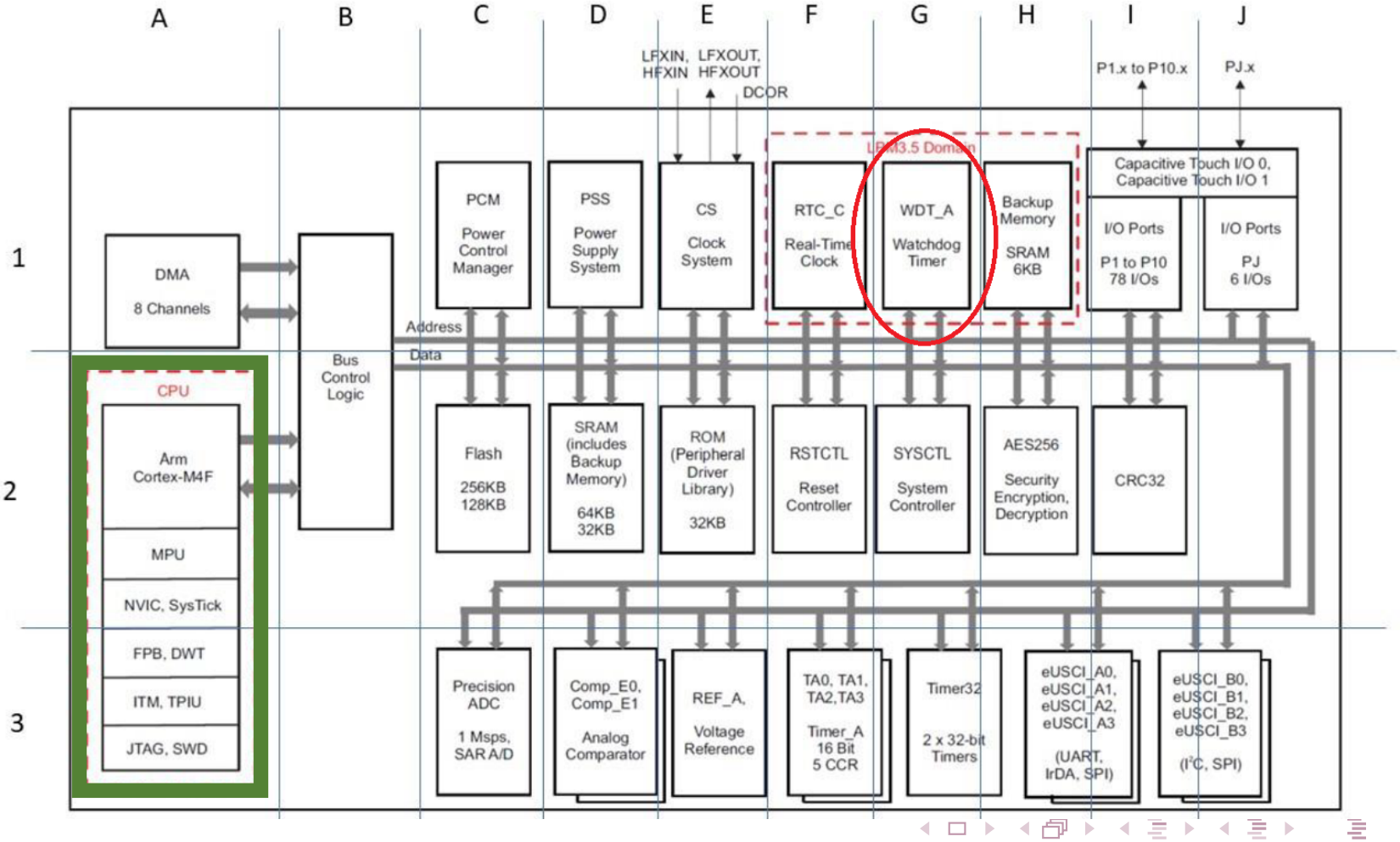
The processor clears the counter for the Watchdog timer to make sure it does not rollover.	The processor has not cleared the counter for the Watchdog timer and the watchdog timer rolls over, which means the processor will reset.

The MSP432 Watchdog Timer(WDT)

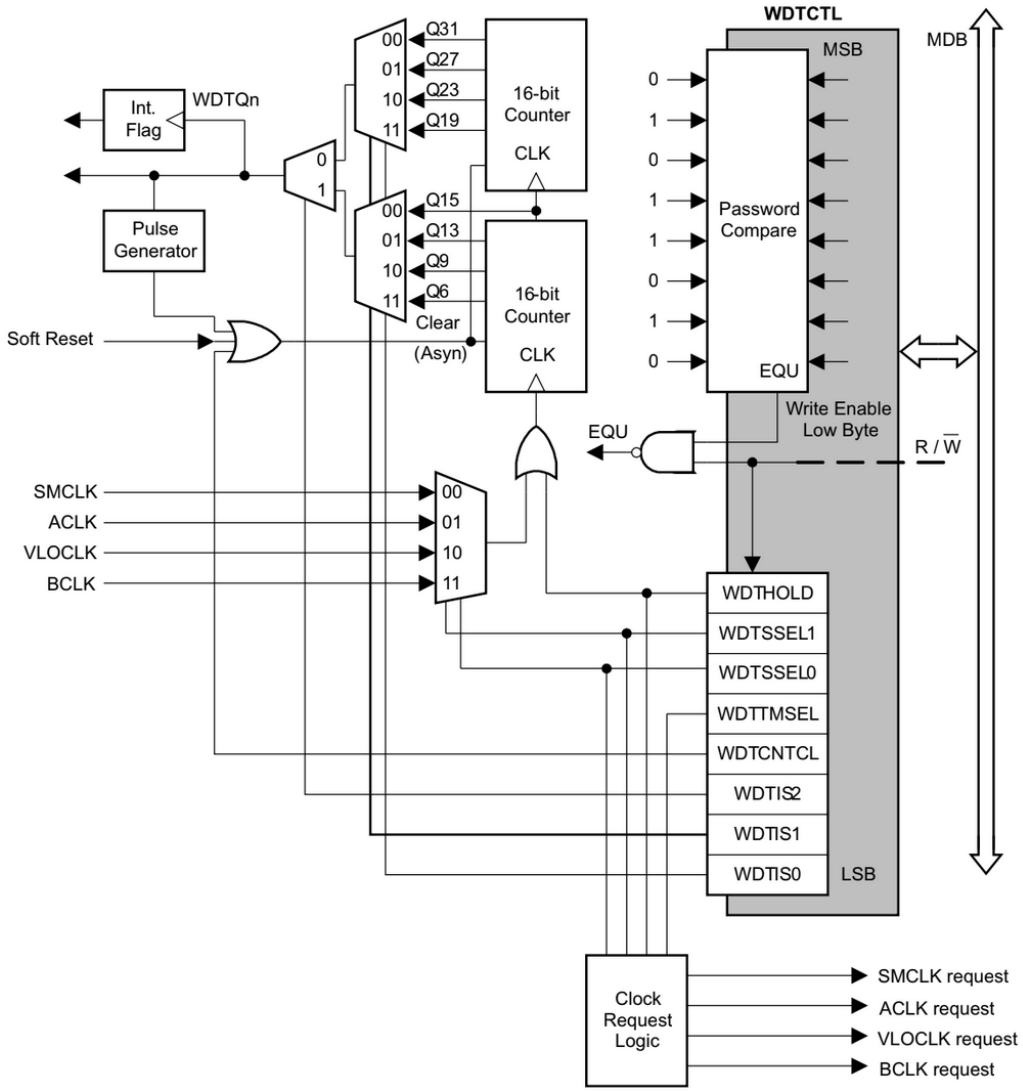
- The WDT is an up-counter
- The WDT increments once per clock cycle, but is reset by certain processor behaviors
- Two 16-bit counters (instead of one 32-bit counter) to simplify the comparator
- Only certain numbers of clock iterations can be used

```
86 #define WDT_A_CLOCKDIVIDER_2G
87 #define WDT_A_CLOCKDIVIDER_128M
88 #define WDT_A_CLOCKDIVIDER_8192K
89 #define WDT_A_CLOCKDIVIDER_512K
90 #define WDT_A_CLOCKDIVIDER_32K
91 #define WDT_A_CLOCKDIVIDER_8192
92 #define WDT_A_CLOCKDIVIDER_512
93 #define WDT_A_CLOCKDIVIDER_64
```

MSP432 Block Diagram



MSP432 Watchdog Circuit Diagram



Initializing an MSP432 Watchdog Timer: 3 Easy Steps

```
void initializeWatchdogTimer() {

 // We can specify resets as either soft-resets or hard-resets. A soft reset allows
 // us to check a different flag to see if the WDT reset the system, but a hard
 // reset is indistinguishable from unplugging the board and plugging it back in.
 SysCtl_setWDTTimeoutResetType(SYSCTL_SOFT_RESET);

 // Initialize the WDT to use a value of 8192000 in its count-up timer
 WDT_A_initWatchdogTimer(WDT_A_CLOCKSOURCE_SMCLK, WDT_A_CLOCKITERATIONS_8192K);

 // In this example, we do not change the clock source of the watchdog timer.
 // By default this clock is the system clock.

 // Since the system clock is 3MHz in this example, the watchdog interval is
 // 8,192,000/3,000,000 ~ 2.73 seconds.
 // So, if we don't reset the watchdog timer within 2.73 s from starting it,
 // it will reset the processor.

 // Start the watchdog timer
 WDT_A_startTimer();
}
```

Testing our Knowledge on a WDT

- Consider the initialization on the previous page
- Suppose the clock is 48 MHz
- How long will it take for the WDT to time out?
  - $8192000 \times (1/48000000) = 171$ ms

See Chapter 17 (Page 756) of the MCU Technical Reference Manual for more details of the MSP432 WDT

Basic Example Watchdog Timer

Let's study the code in the example

Summary

- Brownout reset
  - Beware of low-voltage operation
- Watchdog Timer
  - Periodically clear the timer to indicate normal operation
  - Failure to clear the timer causes a reset
  - "Reset the watchdog timer (clear it) or it will reset you!"
- We've been shutting off the WDT at the beginning of our programs

```
// Stop watchdog timer
// We do this at the beginning of all our programs for now.
// Later we learn more about it.
WDT_A_hold(WDT_A_BASE);
```